

File Compressor and Decompressor Project Report

Objective

The goal of this project is to develop a desktop application that allows users to compress (zip) and decompress (unzip) multiple files through an easy-to-use graphical interface. This tool uses Java's standard `java.util.zip` package for compression and extraction, ensuring no third-party dependencies. The UI is built in JavaFX to provide a responsive and modern user experience.

Tools and Technologies

- **Programming Language:** Java
- **UI Framework:** JavaFX (for desktop GUI)
- **Compression Library:** `java.util.zip` `ZipOutputStream` for compression, `ZipInputStream` for decompression.
- **IDE** Eclipse
- **Build Tool:** Eclipse export for runnable JAR
- **Operating System:** Platform-independent (tested on Windows)

Implementation Approach

- **User Interface:**
Created using JavaFX, the UI includes buttons for compressing and decompressing files, a file chooser dialog for selecting input/output files/folders, and a status area for displaying operation results.
- **Compression:**
Multiple files can be selected and compressed into a single ZIP file using `ZipOutputStream`. Each file is wrapped as a `ZipEntry` and written sequentially to the output ZIP.

- **Decompression:**
ZIP files can be opened and extracted using `ZipInputStream`. The application iterates over each `ZipEntry` to recreate the original files in the user-specified directory.
- **Progress and Logging:**
The application displays progress updates and logs compression or decompression status messages in the UI for enhanced user feedback.
- **Exception Handling:**
The program includes error-handling mechanisms for file IO errors, user cancellations, and invalid operations.

Conclusion

This project successfully provides a lightweight, standalone desktop tool for efficient file compression and decompression. It leverages JavaFX for the UI and the built-in `java.util.zip` package for archiving functionality. The tool supports multiple file selection, delivers real-time feedback, and simplifies file management tasks for users without the need for additional software.